Motivation

Decline of coral reefs under climate change will affect individuals all over the world who rely on their local reefs.

Many developing countries rely on reefs for **sustenance** and **income from tourism and fisheries**, while many benefit from the **coastal protection** provided by reef structures' attenuation of wave action.

Declining reef cover **diminishes these services** and **disproportionately affects countries whose GDP is most linked to** reefs. Here, we aim to **forecast this decline**.

Specific motivation

Presenting at the International Coral Reef Symposium (ICRS) in Auckland, New Zealand in July 2026

Questions

Depreciation

1. Cover as an approximate method for predicting decline in a) tourism, b) fisheries, c) coastal protection?
2. Existing efforts?

Data

1. Best use of your datasets (tourism, coastal protection)

Modelling Global Coral Reef Fisheries states “National level reporting to the FAO has been estimated to capture only 5-25% of coral reef fisheries and may also be failing to capture overall trends (Zeller et al. 2007, Pauly and Zeller 2014)” (Page 3)

2. How to transform your mass estimates into valuation?

Tourism

~ 70 million annual tourist trips
~ $36 billion annual expenditure

Fisheries

Millions' main protein source
Support ~6 million reef fishers
~ $6.8 billion in 2011 revenue
Women represent ~50% workforce

Shoreline protection

Protects ~150,000 km of shoreline
Reduces expected storm damage by > $4 billion annually

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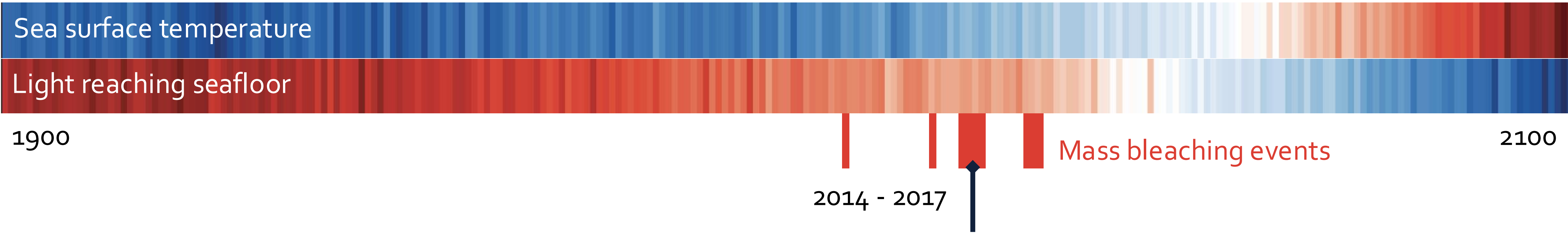
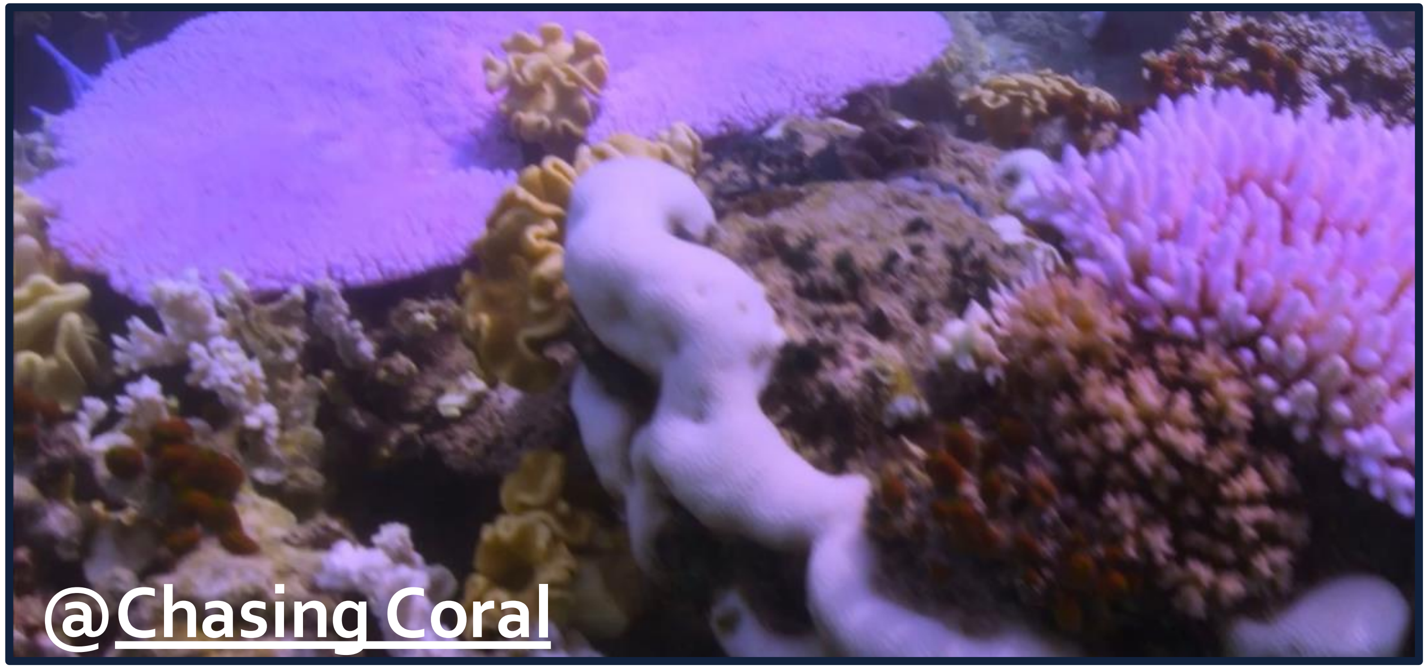
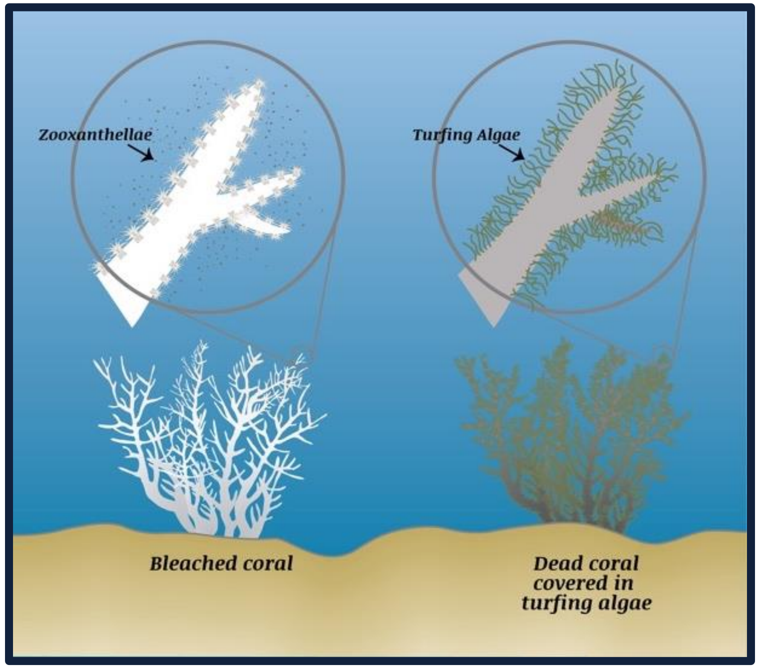

Sea surface temperature
Light reaching seafloor
1900
2100
Mass bleaching events
2014 - 2017


@Chasing Coral


Zooxanthellae
Turfing Algae
Bleached coral
Dead coral covered in turfing algae

Coral cover projection

Using environmental features and in situ observations using a Hierarchical Bayesian Beta model trained on historic conditions and projected on forecasted SSP scenarios.

Sully et al (2022)

Reef associated revenue

Granular attribution of revenue to nations of tourism, fisheries, coastal protection. These datasets are created with various methodologies.

Spalding et al. (2017)
Spalding et al. (2021)
Burke & Spalding (2022)

Depreciation models

Chen et al. (2014)

Quantitative meta-analysis gives a percentage decline in reef associated revenue as a function of percentage change in coral cover. This linear approximation is extended to model diminishing decline and tipping points.

=

Spatially-explicit projections of **climate impacts** on reef-associated **revenue**

Tourism

Mapping the global value and distribution of coral reef tourism, Spalding et al. (2017) combined multiple datasets to **estimate touristic revenue**

Considered **direct** (on-reef) and **indirect** (reef adjacent) effects

Fisheries

Modelling Global Coral Reef Fisheries Spalding et al. (2021) estimates annual mass of catch associated with reefs per km

Sea Around Us provides annual time series of the dollar value of taxa, by territory

Shoreline protection

Shoreline protection by the world's coral reefs: Mapping the benefits to people, assets, and infrastructure, Burke & Spalding (2022) estimated damages to coastal infrastructure avoided due to reefs per decade

Tourism

Global value **USD 36 billion annually**

Very high-value reefs found in 70% of reef jurisdictions

~70% of reefs had **zero touristic-attributable revenue**

Jurisdictional-level tourism values
International & domestic arrivals/spending

'Photo User Days' + Hotel rooms

Spatially-distributed tourism values

Non-urban + >2km from coast; <30km from reefs

Coastal, non-urban tourism values near reefs

Total revenue attributable to reefs

Reef-adjacent tourism | **On-reef tourism**

Value attribution to reefs

***Reef-adjacent** value distributed proportional to reef proximity*

***On-reef** values distributed based on proximity & quantity of Photo User Days & dive sites*

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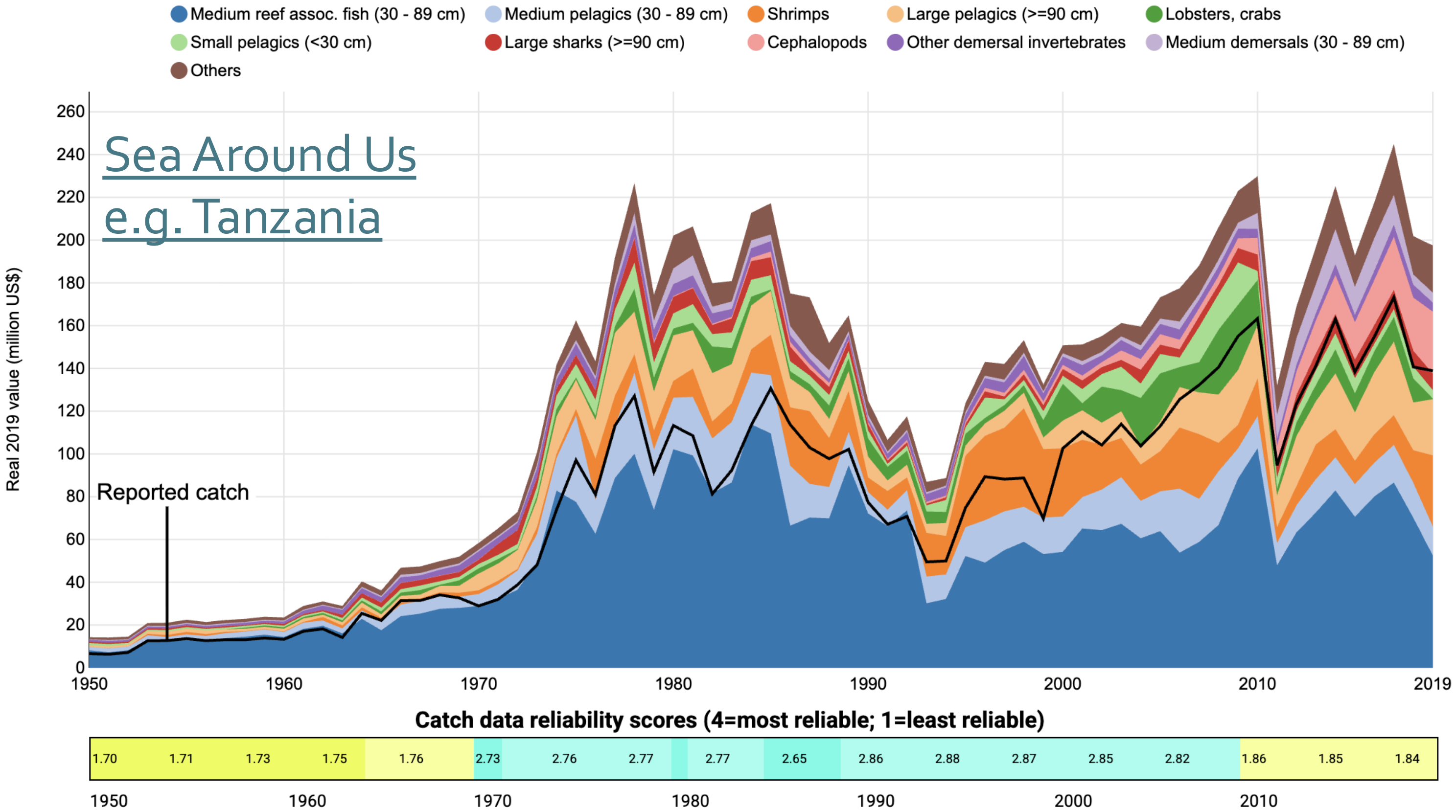
Fisheries

[Sea Around Us](#) combines official reported data from [FAO's FishStat database](#) with reconstructed estimates of unreported data

Data: Dollar value by taxa, functional groups, fishing sector etc.
Considerations: Poor reporting of subsistence take, non-economic value of fisheries



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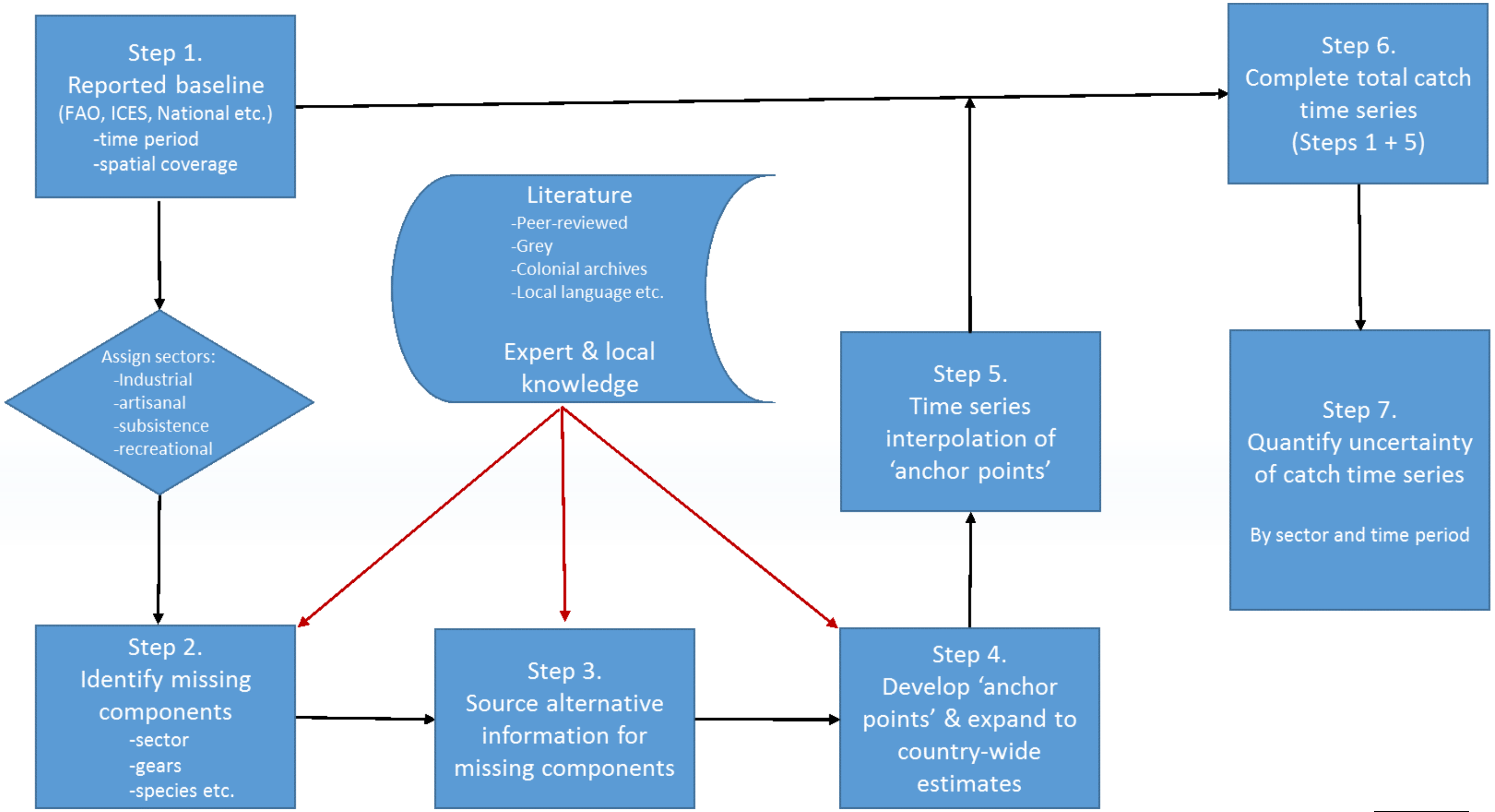
Fisheries

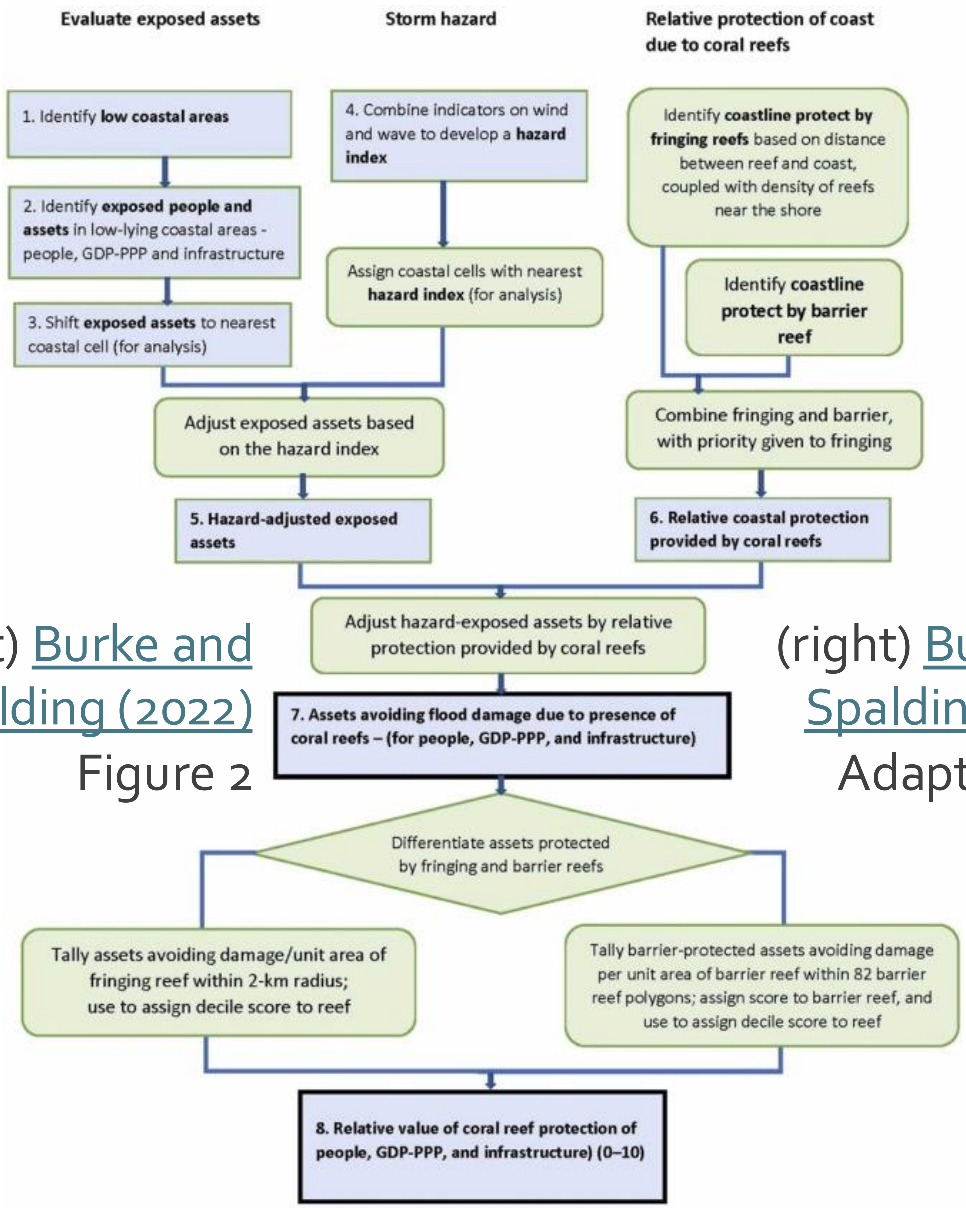
Sea Around Us
Catch Reconstruction and Methods
Figure 2

Sea Around Us combines official reported data from FAO's FishStat database with reconstructed estimates of unreported data

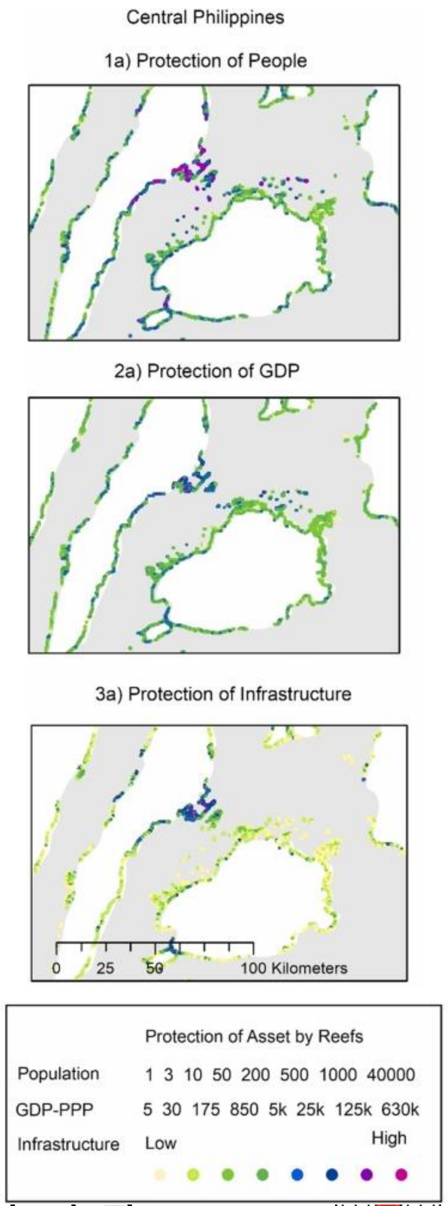


Shoreline Protection

Burke and Spalding (2022) used proximity-based models to estimate the number of people and value of infrastructure protected by attenuation of wave action by reefs



(right) Burke and Spalding (2022) Figure 2

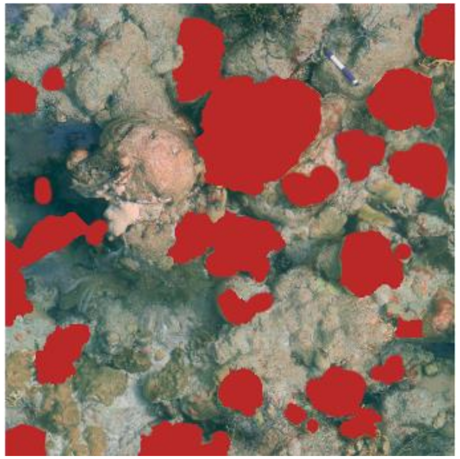


(right) Burke and Spalding (2022) Adapted from Figure 4

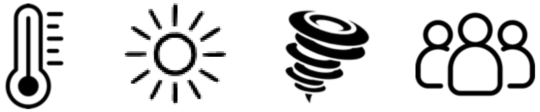
Coral cover projection

Inspired by Sully et al. (2022)

Predictor



Parameters



Model

$$o_i \sim Beta(\alpha_i, \beta_i)$$

$$E(o_i) = \frac{\alpha_i}{\alpha_i + \beta_i} = p_i$$

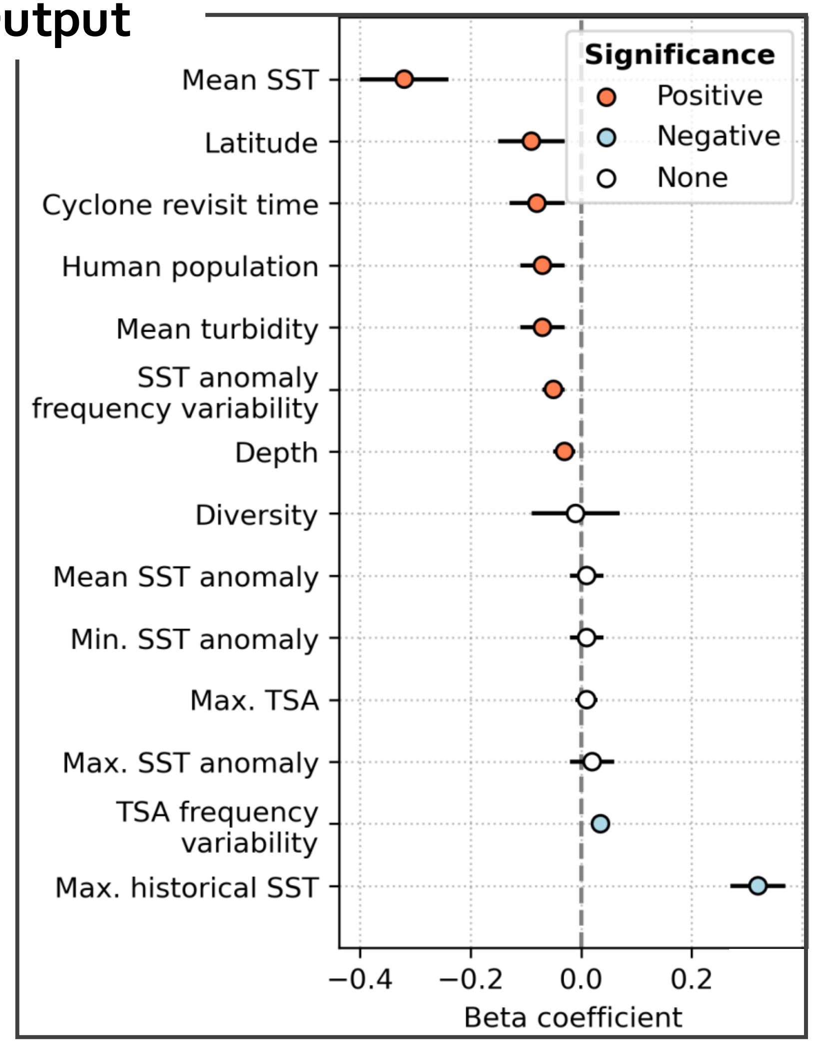
$$Var(o_i) = p_i \cdot \frac{(1 - p_i)}{(\emptyset + 1)}$$

$$logit(p_i) = \sum_n \beta_n x_{i,n} + a_{site,ID(i)}$$

$$a_{site} \sim \mathcal{N}(r_{r,s}, \sigma_{r,s}^2)$$

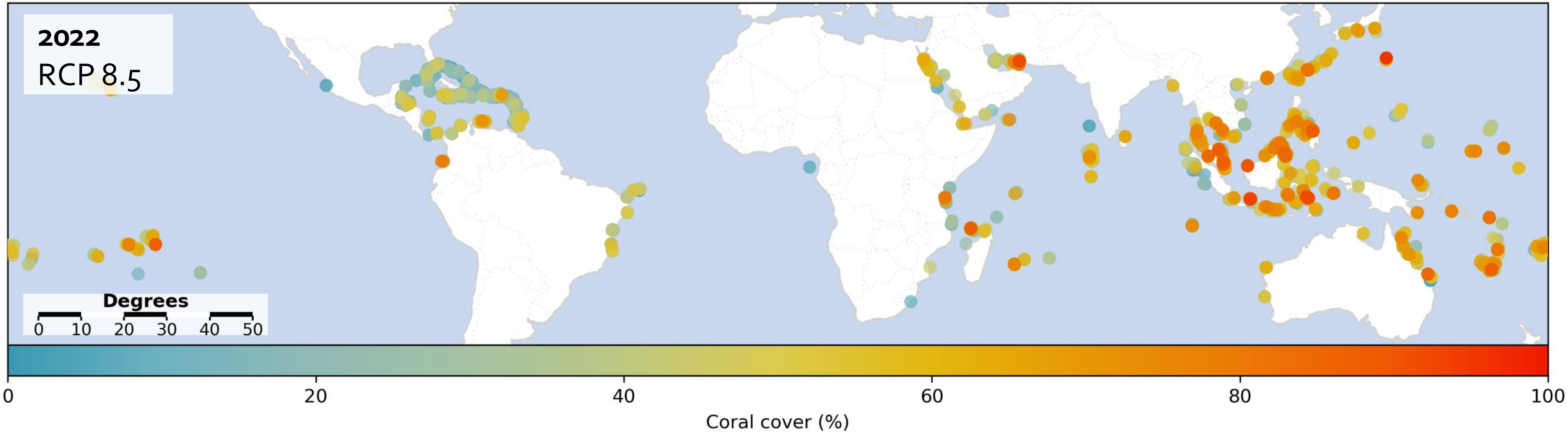
$$r_{region} \sim \mathcal{N}(\mu + \beta_{diversity} \cdot d_r, \sigma_r^2)$$

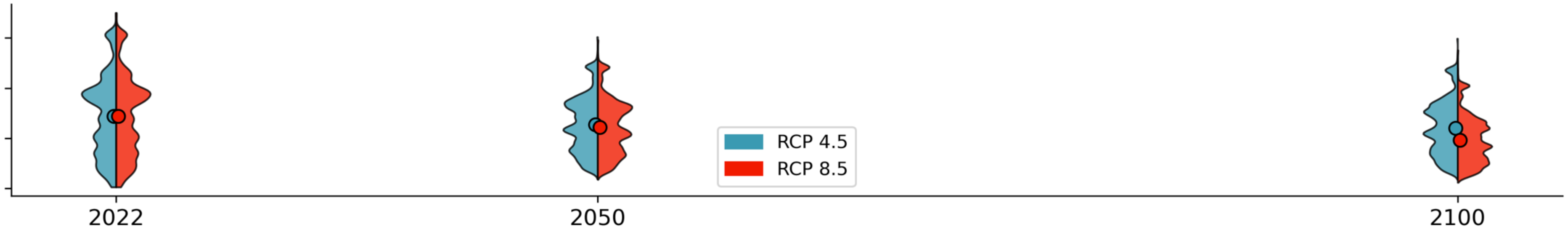
Output



Coral cover projection

Inspired by Sully et al. (2022)





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Depreciation Models

Chen et al. (2014)

Meta-analysis of **72 regional studies** which quantify reef value as function of coral cover via:

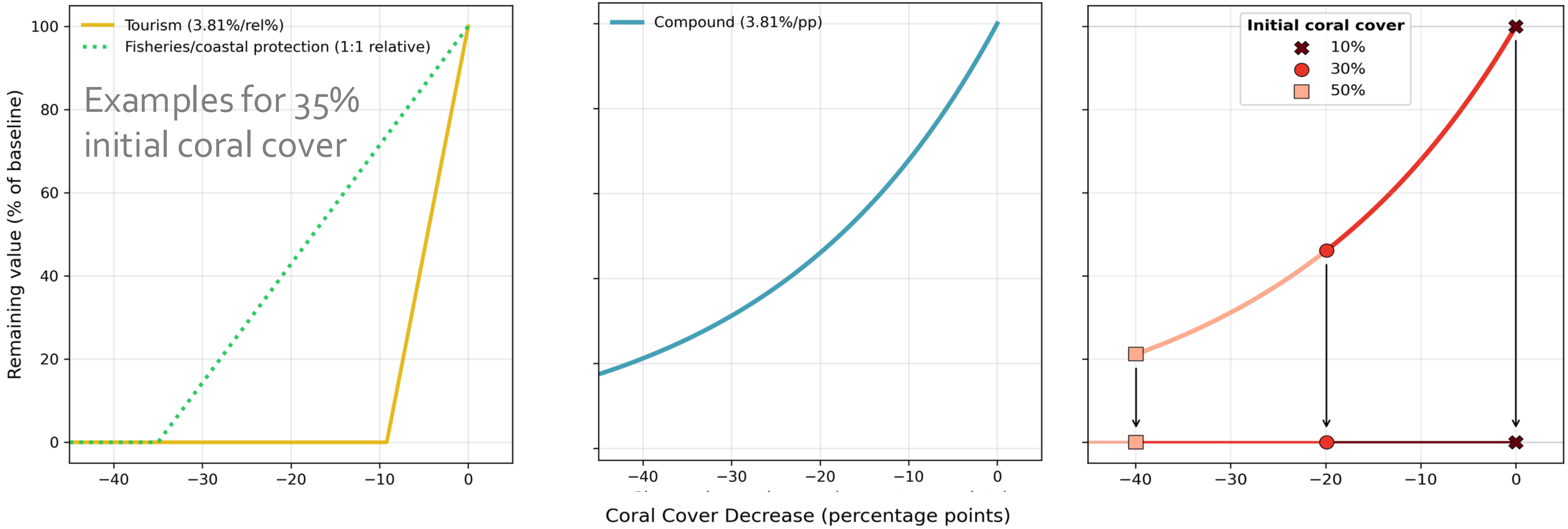
- **Contingent valuation**
- **Travel cost**
- **Hedonic price**

Non-linear terms in coral cover; linear in all other parameters

Author- and observation-level error terms

$Value_{i,j} = \alpha +$	Intercept
$\beta_1 \cdot Cover_{i,j} + \beta_1 \cdot Cover_{i,j}^2 +$	Quadratic in coral cover
$\beta_3 \cdot GDP_{i,j} + \beta_4 \cdot Visitor_{i,j} + \beta_5 \cdot TE_{i,j} +$	TE is per-visitor expenditure
$\sum_{k=6,7} \beta_k \cdot region_{i,j} +$	Ocean basin term
$\beta_8 \cdot published_{i,j} +$	Publication status
$\beta_9 \cdot CVM_{i,j} +$	Contingent value method mask
$u_j + \varepsilon_{i,j}$	Author and observation-level errors

Depreciation Models



Linear

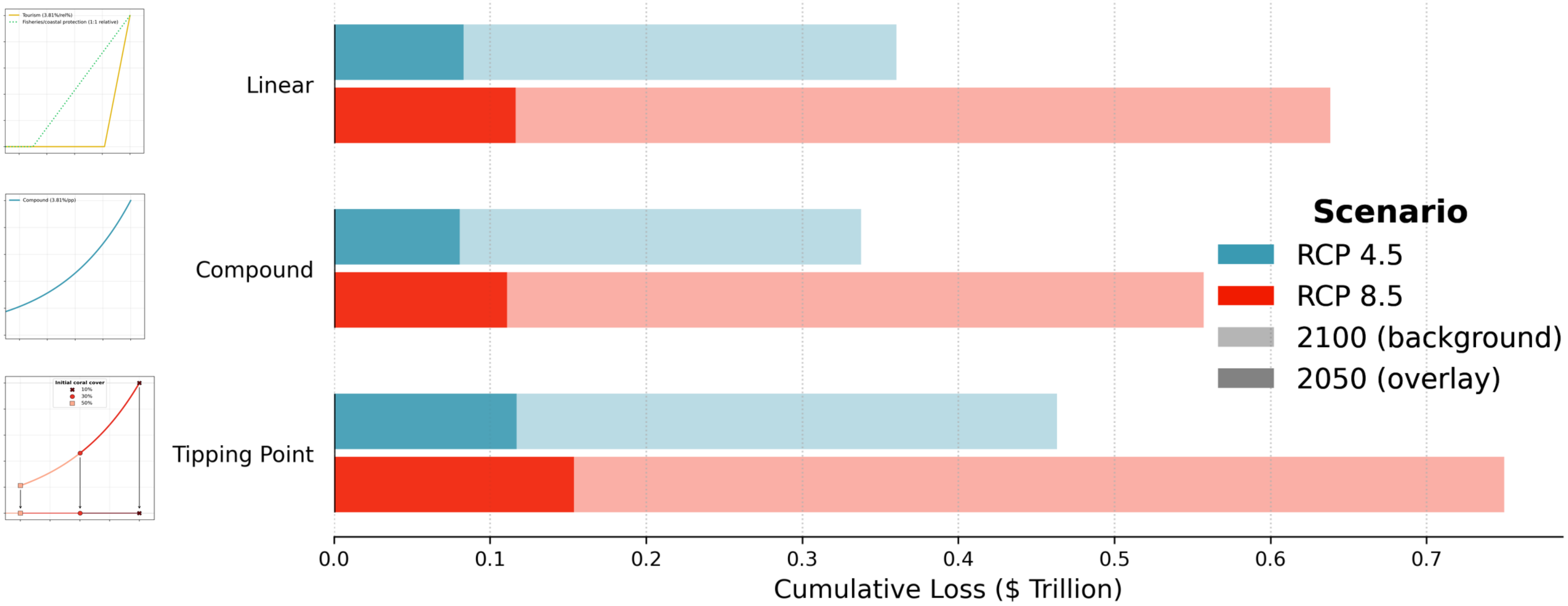
3.81% ↓ V_0 per 1% ↓ coral cover

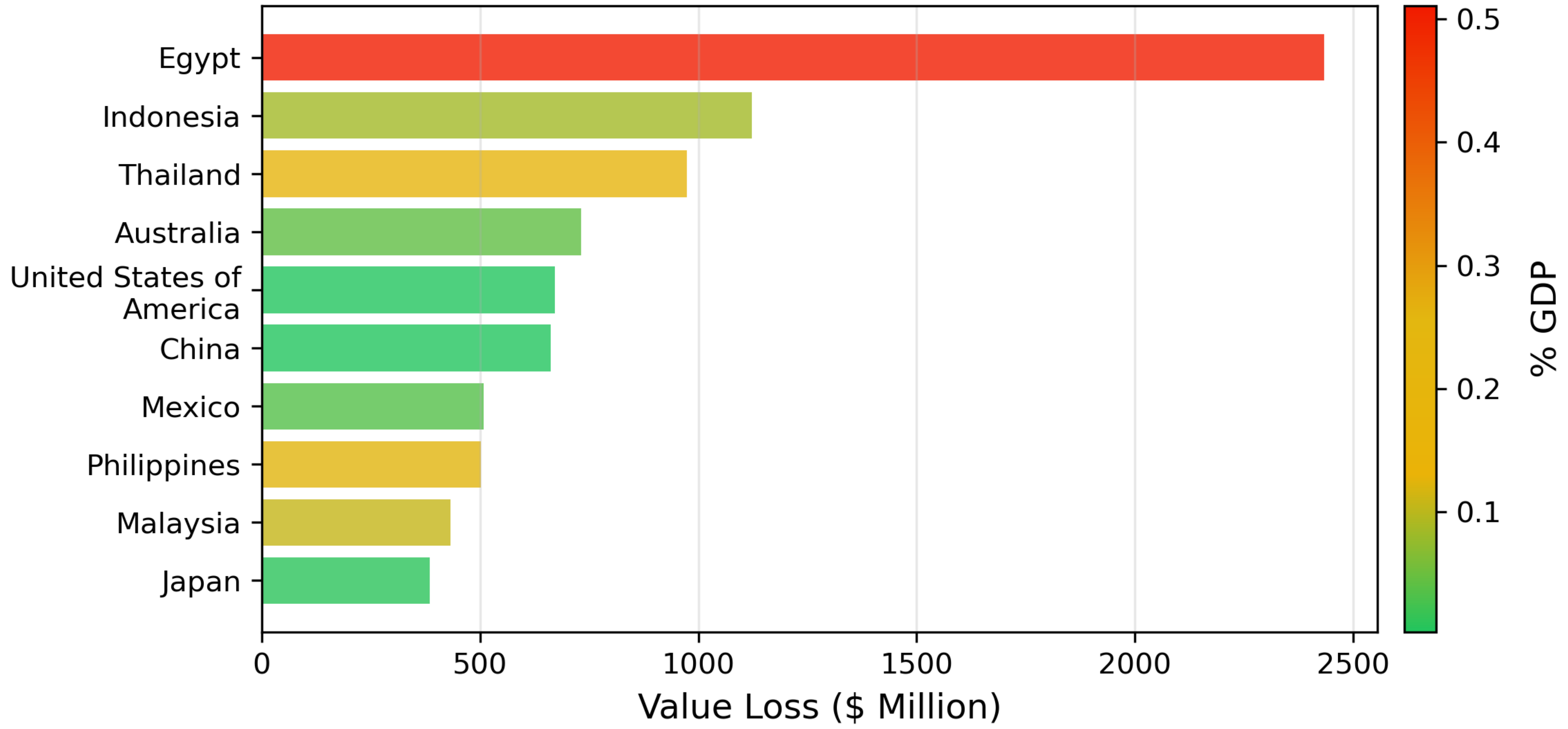
Compound

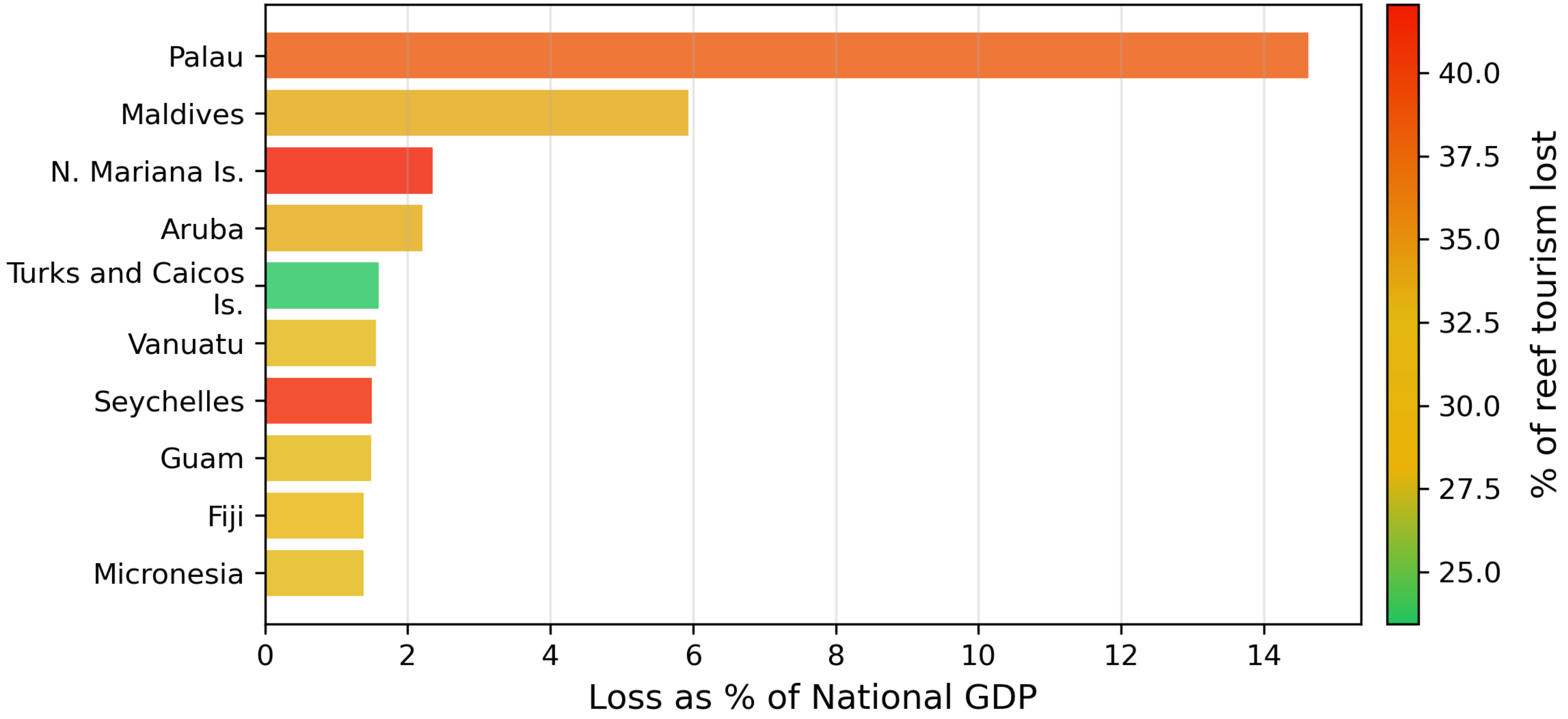
3.81% ↓ V per 1% ↓ coral cover

Tipping point

Compound model + sudden ↓ once 10% coral cover reached


Linear
Compound
Tipping Point
Initial coral cover
10%
30%
50%
Scenario
RCP 4.5
RCP 8.5
2100 (background)
2050 (overlay)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Cumulative Loss ($ Trillion)


Egypt
Indonesia
Thailand
Australia
United States of America
China
Mexico
Philippines
Malaysia
Japan
0
500
1000
1500
2000
2500
Value Loss ($ Million)
0.5
0.4
0.3
0.2
0.1
% GDP



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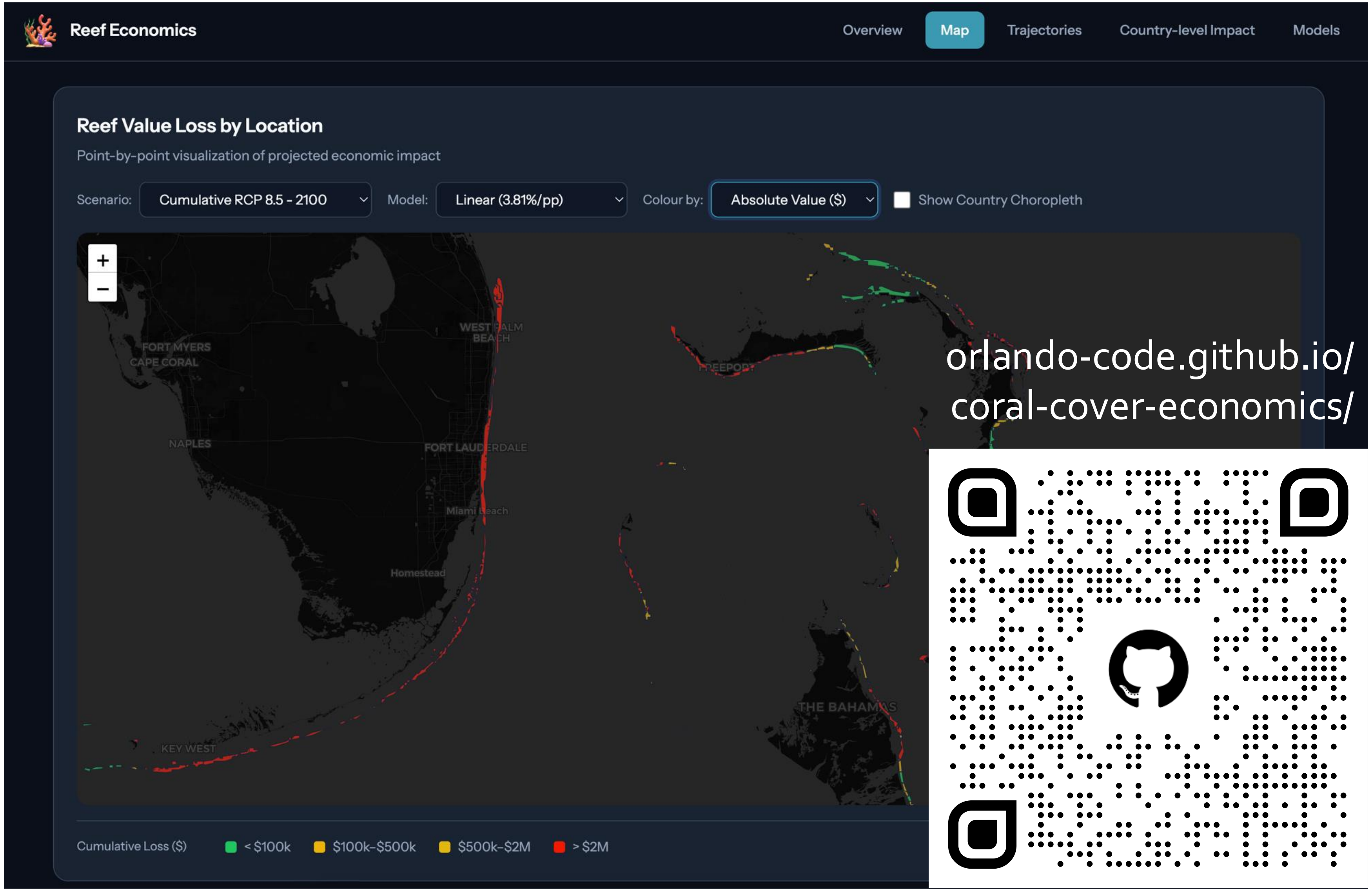
Further work

Improve predictive cover model via additional inputs and robust validation

Validate projected decreases using historical cover and economics timeseries

Contextualise **non-economic impacts** of reef cover decline



orlando-code.github.io/
coral-cover-economics/